

**Definition 15.1**

Let  $G$  be a group and  $X$  be a set. Given a function

$$\mu: G \times X \rightarrow X$$

denote  $g \cdot x := \mu(g, x)$ .

We say that  $\mu$  is a *group action* of  $G$  on the set  $X$  if the following conditions are satisfied:

- 1)  $(gh) \cdot x = g \cdot (h \cdot x)$  for any  $g, h \in G$  and  $x \in X$ .
- 2)  $e \cdot x = x$  for any  $x \in X$ .

### Definition 15.2

Let  $\mu: G \times X \rightarrow X$  be a group action.

- The *orbit* of an element  $x \in X$  is the subset of  $X$  given by

$$\text{Orb}(x) = \{gx \mid g \in G\}$$

- The *stabilizer* of an element  $x \in X$  is the subset of  $G$  given by

$$\text{Stab}(x) = \{g \in G \mid gx = x\}$$

### Theorem 15.3

Let  $\mu: G \times X \rightarrow X$  be a group action and let  $x, y \in X$ . Then:

- 1)  $x \in \text{Orb}(x)$ .
- 2) Either  $\text{Orb}(x) = \text{Orb}(y)$  or  $\text{Orb}(x) \cap \text{Orb}(y) = \emptyset$ .
- 3)  $\text{Orb}(x) = \text{Orb}(y)$  if and only if  $y = gx$  for some  $g \in G$ .

### Corollary 15.4

If  $\mu: G \times X \rightarrow X$  is a group action and  $X$  is a finite set, then

$$|X| = |\text{Orb}(x_1)| + |\text{Orb}(x_2)| + \cdots + |\text{Orb}(x_m)|$$

where  $\text{Orb}(x_1), \text{Orb}(x_2), \dots, \text{Orb}(x_m)$  are all different orbits of the action.

### Theorem 15.5

Let  $\mu: G \times X \rightarrow X$  be a group action and let  $x \in X$ .

- 1)  $\text{Stab}(x)$  is a subgroup of  $G$ .
- 2) If  $y = gx$  then  $\text{Stab}(y) = g \text{Stab}(x) g^{-1}$ .
- 3) If  $G$  is a finite group then  $|G| = |\text{Orb}(x)| \cdot |\text{Stab}(x)|$

### Theorem 15.6 (Cauchy Theorem)

If  $G$  is a finite group and  $p$  is a prime that divides  $|G|$  then there exists an element of order  $p$  in  $G$ .

**Definition 15.7**

If  $p$  is a prime number then a  $p$ -group is a finite group of order  $p^r$  for some  $r \geq 0$ .

**Corollary 15.8**

A finite group  $G$  is a  $p$ -group if and only if the order of every element of  $G$  is a power of  $p$

**Theorem 15.9**

If  $G$  is a  $p$ -group then there exists an element  $a \in G$  such that  $a \neq e$  and  $ag = ga$  for all  $g \in G$ .